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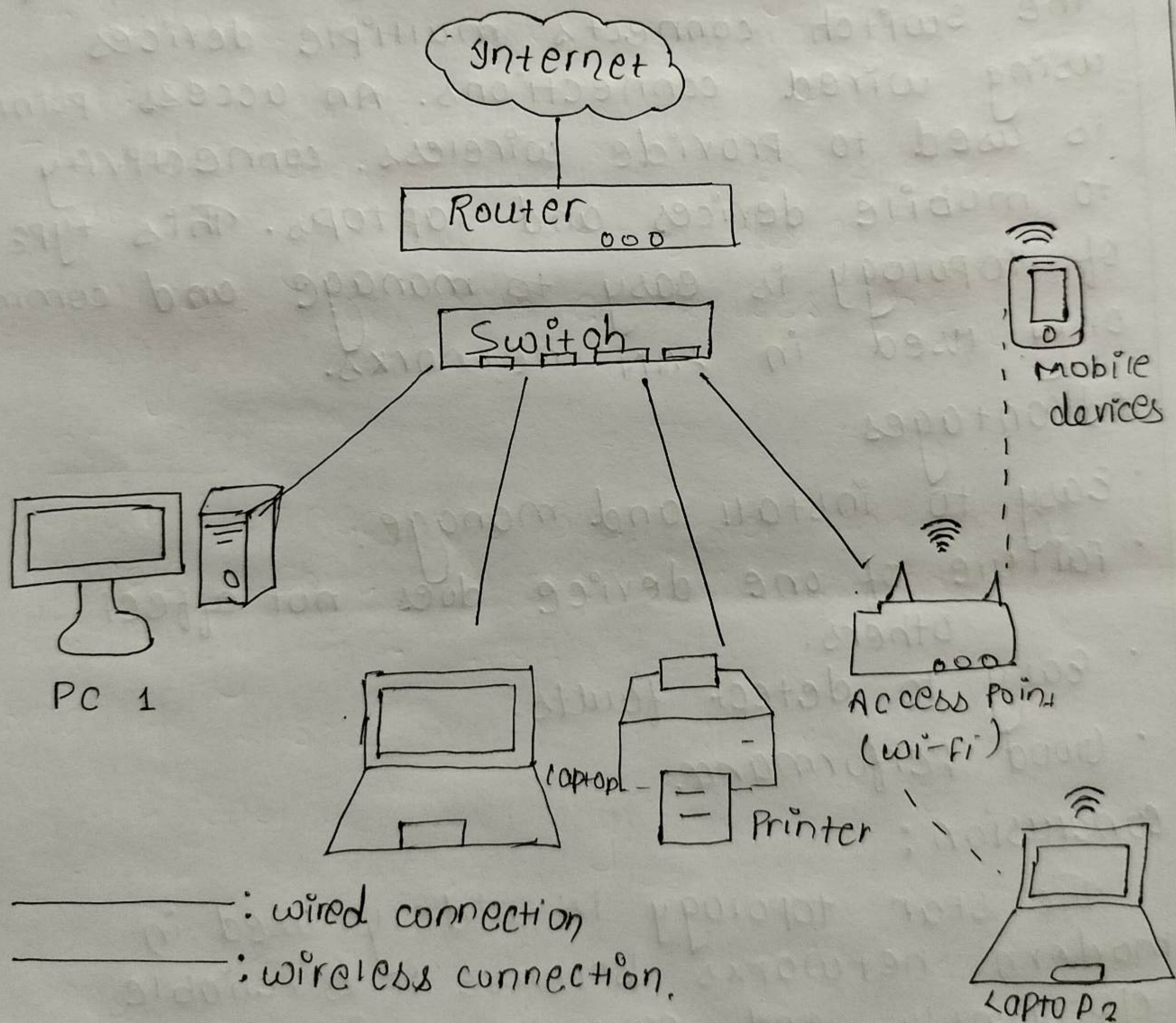
1. Draw the network topology.
2. Identify network devices.
3. Analyze connection types.
4. Measure network speed.
5. Find system network details.
6. Explain observations.

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organisation (e.g. star, bus, mesh).
- clearly label all components such as Router, switch, access point and end devices (PCs, laptops, etc).

Network Topology (star topology):

Aim: To study and draw the LAN network topology used in our organisation.



components used:

- Router.
- Switch.
- Access Point (wi-fi)
- PC
- Laptop
- Printer.
- mobile devices.

The network topology used is Star Topology all devices are connected to a central device such as a switch or router. The router connects the network to the internet, and the switch connects multiple devices using wired connections. An access point is used to provide wireless connectivity to mobile devices and laptops. This type of topology is easy to manage and commonly used in LAN networks.

Advantages.

- Easy to install and manage.
- Failure of one device does not affect others.
- Easy to detect faults.
- Good Performance.

Conclusion:

The star topology is widely used in modern networks because it is reliable, efficiently and easy to maintain.

2. Identification of Network devices in LAN:

1. Introduction:

A local Area Network (LAN) consists of multiple interconnected devices that enable communication, data sharing, and resource utilization within a limited geographical area such as a home, office or campus. Various networking devices are used to establish, manages and maintain this communication efficiently.

2. List of Networking devices in LAN:

2.1 Router:

Function:

- connect different networks (e.g. LAN to the internet)
- Routes data packets based on IP address.
- Provides internet access to multiple devices
- often includes built-in firewall and DHCP services.

2.2 Switch:

Function:

- connects multiple devices within a LAN
- Transfers data using MAC addresses
- Reduces network collisions by sending data only to the intended device.
- Improves network efficiency.

2.3 HUB

Function:

- connects multiple devices in a network (basic device)
- Broadcasts data to all connected devices.
- works on Physical layer (layer 1).
- Less efficient compared to switches.

2.4 Bridge

Function:

- connects two LAN segments.
- Filters traffic based on MAC addresses
- Reduces unnecessary data transmission.
- works at Data link - layer (layer 2).

2.5 Repeater

Function:

- Regenerates and amplifies weak signals.
- Extends the transmission distance of a networks.
- works at Physical layer.

2.6 Access Point

Function:

- Provides wireless connectivity to devices.
- connects wireless devices to a wired LAN.
- commonly used in wi-fi networks

2.7 Modem

Functions:

- converts digital systems to analog and vice versa.
- connects a network to an internet service Provider (ISP).
- Required for internet access in many setups.

2.8 Network Interface Card (NIC):

Functions:

- Allows a device to connect to a network
- Provides a unique MAC address.
- can be wired (Ethernet) or wireless (Wi-Fi)

2.9 Firewall (Hardware / Software)

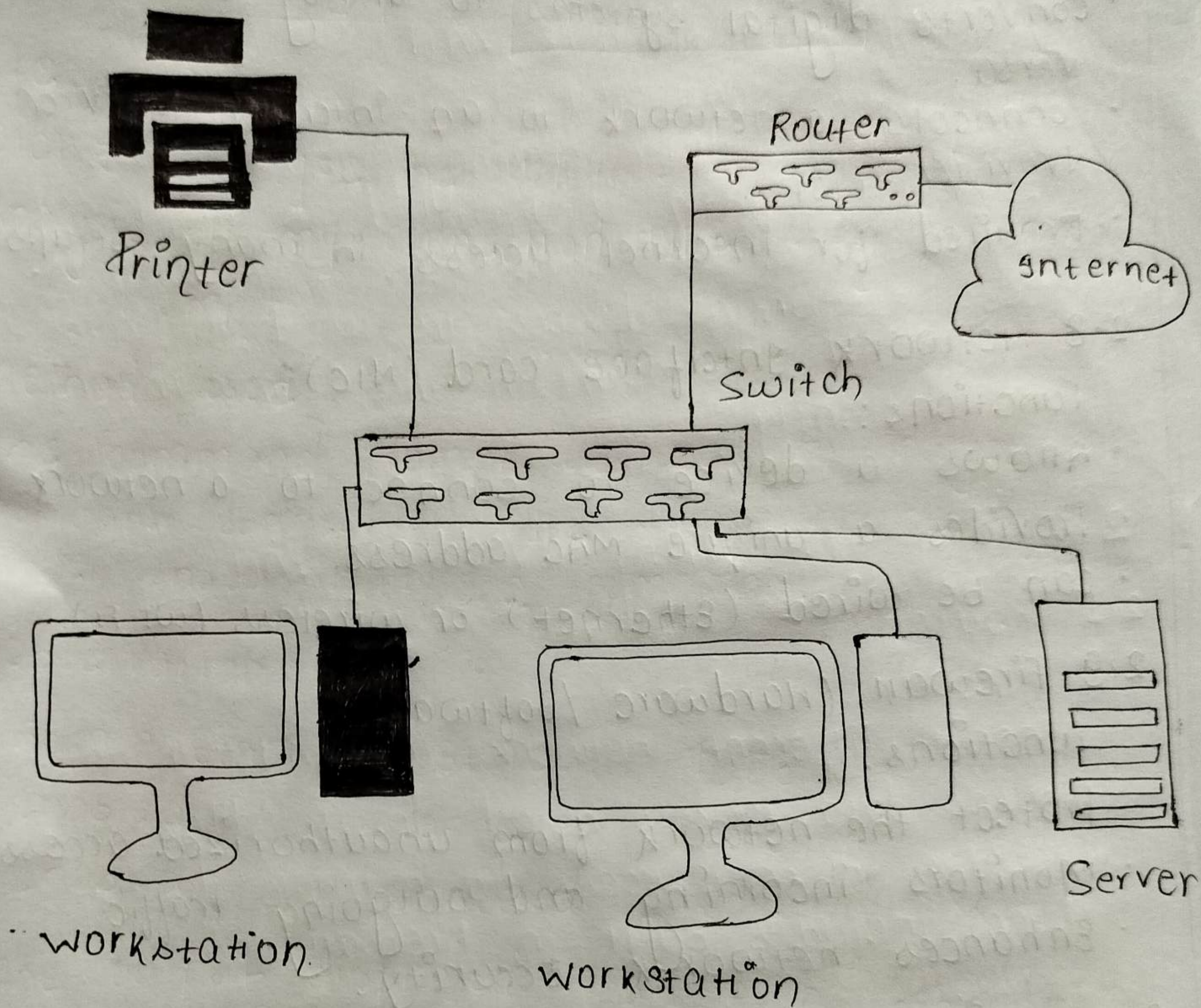
Functions:

- Protect the network from unauthorized access.
- Monitors incoming and outgoing traffic.
- Enhances network security.

Conclusion:

Networking devices play a crucial role in the functioning of a LAN. Each device has a specific purpose, from connecting devices and routing data to ensuring security and efficient communication. Proper understanding and configuration of these devices are essential for maintaining a reliable and secure network.

LAN Network Diagram:



3. Analyze connection types:

Specify whether connections are:

- wired (Ethernet)
- wireless (Wi-Fi)

How to analyze connection type?

way 1: check physically

→ If Ethernet cable is connected.

- you will see a LAN cable (RJ45) connected to your device.

- connection type = wired (Ethernet)

→ If no cable, using Wi-Fi

- no cable connected.

- you see Wi-Fi icon on screen

- connection type = wireless (Wi-Fi)

way 2: check in your laptop (Windows)

1. Go to settings.

2. Click network and internet.

3. See: If it shows Ethernet → wired

If it shows Wi-Fi → wireless

way 3: using cmd

1) open command prompt.

- Press windows + R.

- type cmd & press enter.

Step 2: Type this command

ipconfig

or

netsh interface
show interface

Press enter

Step 3: Analyze the output.

Analysis of my connection type (using command Prompt)

Method used:

The command Prompt was used to analyze the network connection type using the netsh interface show interface command.

Observation:

Admin State	State	Type	Interface Name
Enabled	connected	Dedicated	Ethernet
Enabled	connected	Dedicated	Wi-Fi

The output displayed that both "Ethernet" and "Wi-Fi" interfaces were in a connected state, while the de

Result:

The system is primarily connected using a wired (Ethernet) connection, as the Ethernet interface is active.

Conclusion:

The wired connection provides better stability and higher speed compared to wireless connection making

- mention where each type is used.

In a local area Network (LAN) devices can be connected using either wired or wireless communication methods. Each type of connection has its own advantages, limitations and specific use cases.

Types of connections :-

1) wired connection (ethernet)

A wired connection uses physical cables (ethernet cable) to connect devices within a network.

Features:

- high speed and stable connection.
- low interfaces.
- more secure compared to wireless.

where it is used?

- office and companies.
- computer labs.
- servers and data centers.
- gaming setup requiring stable internet.

2) wireless connection (wi-fi)

A wireless connection uses radio waves to connect devices without physical cables.

Features:

- easy to install and use.
- provides mobility.
- can connect multiple devices easily.

where it is used ?

- Homes and Hostels.
- cafes and public places.
- Mobile devices (phones, tablets, laptops)
- smart devices (IoT)

4. Measure Network Speed :

- Find the link speed of :
 - wired network (if available)
 - wireless network

4.1 objective

To measure the link speed of the wireless (wi-fi) network using command prompt.

4.2 Method used.

The command prompt was used to obtain detailed information about the wireless network including the transmission (T_x) and reception (R_x) link speed.

4.3 command used

netsh wlan show interface.

4.4 observation.

After executing the command, the following details were observed.

- Name : wi-fi
- State : connected
- SSID : Panklavi
- Receive rate (Mbps): 144.4
- Transmit rate (Mbps): 144.4

4.5 Result :

The wireless network link speed is

- Receive speed = ~~300 Mbps~~ 144.4 Mbps
- Transmit speed = 144.4 Mbps

Conclusion :

The system is connected using a wireless network. The command prompt was used to measure the wireless network speed using the command `netsh wlan show interfaces`. The receive rate was 144.4 Mbps. and the transmit rate was 144.4 Mbps. This indicating stable wireless connection suitable for normal usage.

• Mention the tools / commands used (e.g. Task manager, netsh; `netsh wlan show interfaces` or, etc).

To analyze the network connection type and measure the network speed, various system tools and command-line utilities were used. These tools help in obtaining accurate and detailed information about the network.

Tools used :

1) command prompt (cmd).

Purpose :

- To analyze network interfaces.
- To check connection type (wired / wireless)
- To measure network speed.

2) Task manager :

Purpose :

- To monitor real-time network usage.
- To observe data transfer speed.

commands used:

1. `ipconfig`

function:

- Display IP configuration details.
- helps identify whether connection is ethernet or wi-fi.

2. `netsh interface show interface`

function:

- Shows all network interfaces and their status.
- helps determine which connection is active.

3. `netsh wlan show interfaces`

function:

Displays detailed information about the wireless network

shows Receive and Transmit speed (Mbps)

5. Find System Network details:

- Determine and record the following of your computer:
 - MAC address
 - IP Address (IPv4)
- Mention the commands used (eg ipconfig, getmac).

5.1 Objective:

To determine and record the systems network details such as MAC address, IPv4 address and subnet mask using command line tools:

5.2 Method used:

The command prompt was used to retrieve network configuration details using specific commands.

5.3 commands used.

The command used use:

getmac - for mac address.

ipconfig - for ip address.

5.4 observation:

After executing the above commands the following details were obtained:

- MAC Address : 50-BB-B5-7B-10-D2
- IPv4 Address : 192.168.1.66
- Subnet Mask : 255.255.255.0

conclusion :

The system network details were successfully obtained using command prompt these details are essential for identifying and communicating with devices in a network.

Reference Photo :

6. Explain observations:

• Briefly explain:

- Difference between wired and wireless speed.
- Role of subnet mask in networking

Difference between wired and wireless speed.
wired and wireless networks differ significantly in terms of speed and performance.

• wired connection (ethernet):

- Provides higher and more stable speed.
- less interference and signal loss.
- suitable for tasks requiring high bandwidth

• wireless connection (wi-fi)

- Provides comparatively lower speed.
- Speed may vary due to distance and.
- Suitable for mobile devices and general usage

conclusion:

wired connections are faster and more reliable, while wireless connections offer flexibility and ease of use.

Role of Subnet Mask in Networking

The subnet mask plays an important role in network communication.

- It divides an IP address in network part and host part.

- Helps in identifying whether a device belongs to the same network or not.
- Enables efficient routing of data within a network.
- Reduces network congestion by organising devices into subnets.

Example:

If the IP address is 192.168.1.5 and subnet mask is 255.255.255.0,

- Network Portion $\rightarrow$ 192.168.1.
- Host Portion $\rightarrow$ 5.

Conclusion:

The analysis shows that while wired networks provide better speed and stability, wireless networks are more convenient. Additionally, subnet masks are crucial for efficient data transmission and network management.